Foundations of Softmax function

[Softmax function]

$$\mathrm{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

1.0 Content Block

σ (0 , 10) := σ 1 (0 , 10) = (1 / (1 + e 10) , e 10 / (1 + e 10)) ≈ (0.00005 , 0.99995) {\displaystyle \sigma (0,\,10):=\sigma _{1}(0,\,10)=\left(1/\left(1+e^{10}\right),\,e^{10}/\left(1+e^{10}\right)\right)\approx (0.00005,\,0.99995)}

2.0 Content Block

A value proportional to the reciprocal of β is sometimes referred to as the temperature: β = 1 / k T {\textstyle \beta =1/kT} , where k is typically 1 or the Boltzmann constant and T is the temperature. A higher temperature results in a more uniform output distribution (i.e. with higher entropy; it is "more random"), while a lower temperature results in a sharper output distribution, with one value dominating. In some fields, the base is fixed, corresponding to a fixed scale, while in others the parameter β (or T) is varied. The softmax function is a multiple-variable generalization of the logistic function.

3.0 Content Block

Mathematical properties Geometrically the softmax function maps the Euclidean space R K {\displaystyle \mathbb {R} ^{K}} to the boundary of the standard (K − 1) {\displaystyle (K-1)} -simplex, cutting the dimension by one (the range is a (K − 1) {\displaystyle (K-1)} -dimensional simplex in K {\displaystyle K} -dimensional space), due to the linear constraint that all output sum to 1 meaning it lies on a hyperplane. Along the main diagonal (x , x , … , x) , {\displaystyle (x,\,x,\,\dots ,\,x),} softmax is just the uniform distribution on outputs, (1 / n , … , 1 / n) {\displaystyle (1/n,\dots ,1/n)} : equal scores yield equal probabilities. More generally, softmax is invariant under translation by the same value in each coordinate: adding c = (c , … , c) {\displaystyle \mathbf {c} =(c,\,\dots ,\,c)} to the inputs z {\displaystyle \mathbf {z} } yields σ (z + c) = σ (z) {\displaystyle \sigma (\mathbf {z} +\mathbf {c})=\sigma (\mathbf {z})} , because it multiplies each exponent by the same factor, e c {\displaystyle e^{c}} (because e z i + c = e z i ⋅ e c {\displaystyle e^{z_{i}+c}=e^{z_{i}}\cdot e^{c}}), so the ratios do not change:

4.0 Content Block

Here, the Kronecker delta is used for simplicity (cf. the derivative of a sigmoid function, being expressed via the function itself). To ensure stable numerical computations subtracting the maximum value from the input tuple is common. This approach, while not altering the output or the derivative theoretically, enhances stability by directly controlling the maximum exponent value computed. If the function is scaled with the parameter β {\displaystyle \beta } , then these expressions must be multiplied by β {\displaystyle \beta } . See multinomial logit for a probability model which uses the softmax activation function.

5.0 Content Block

∂ ∂ q k σ (q , i) = σ (q , i) (δ i k − σ (q , k)) . {\displaystyle {\frac {\partial }{\partial q_{k}}}\sigma ({\textbf {q}},i)=\sigma ({\textbf {q}},i)(\delta _{ik}-\sigma ({\textbf {q}},k)).}

6.0 Content Block

Numerical algorithms The standard softmax is numerically unstable because of large exponentiations. The safe softmax method calculates instead σ (z) i = e β (z i − m) ∑ j = 1 K e β (z j − m) {\displaystyle \sigma (\mathbf {z})_{i}={\frac {e^{\beta (z_{i}-m)}}{\sum _{j=1}^{K}e^{\beta (z_{j}-m)}}}} where m = max i z i {\displaystyle m=\max _{i}z_{i}} is the largest factor involved. Subtracting by it guarantees that the exponentiations result in at most 1. The attention mechanism in Transformers takes three arguments: a "query vector" q {\displaystyle q} , a list of "key vectors" k 1 , … , k N {\displaystyle k_{1},\dots ,k_{N}} , and a list of "value vectors" v 1 , … , v N {\displaystyle v_{1},\dots ,v_{N}} , and outputs a softmax-weighted sum over value vectors: o = ∑ i = 1 N e q T k i − m ∑ j = 1 N e q T k j − m v i {\displaystyle o=\sum _{i=1}^{N}{\frac {e^{q^{T}k_{i}-m}}{\sum _{j=1}^{N}e^{q^{T}k_{j}-m}}}v_{i}} The standard softmax method involves several loops over the inputs, which would be bottlenecked by memory bandwidth. The FlashAttention method is a communication-avoiding algorithm that fuses these operations into a single loop, increasing the arithmetic intensity. It is an online algorithm that computes the following quantities: z i = q T k i m i = max (z 1 , … , z i) =

7.0 Content Block

P t (a) = exp ■ (q t (a) / τ) ∑ i = 1 n exp ■ (q t (i) / τ) , {\displaystyle P_{t}(a)={\frac {\exp(q_{t}(a)/\tau)}{\sum _{i=1}^{n}\exp(q_{t}(i)/\tau)}}{\text{,}}}

8.0 Content Block

The softmax function, also known as softargmax or normalized exponential function, converts a tuple of K real numbers into a probability distribution over K possible outcomes. It is a generalization of the logistic function to multiple dimensions, and is used in multinomial logistic regression. The softmax function is often used as the last activation function of a neural network to normalize the output of a network to a probability distribution over predicted output classes.

9.0 Content Block

The Softmax function is a smooth approximation to the arg max function: the function whose value is the index of a tuple's largest element. The name "softmax" may be misleading. Softmax is not a smooth maximum (that is, a smooth approximation to the maximum function). The term "softmax" is also used for the closely related LogSumExp function, which is a smooth maximum. For this reason, some prefer the more accurate term "softargmax", though the term "softmax" is conventional in machine learning. This section uses the term "softargmax" for clarity. Formally, instead of considering the arg max as a function with categorical output 1 , … , n {\displaystyle 1,\dots ,n} (corresponding to the index), consider the arg max function with one-hot representation of the output (assuming there is a unique maximum arg):

10.0 Content Block

where the action value q t (a) {\displaystyle q_{t}(a)} corresponds to the expected reward of following action a and τ {\displaystyle \tau } is called a temperature parameter (in allusion to statistical mechanics). For high temperatures (τ → ∞ {\displaystyle \tau \to \infty }), all actions have nearly the same probability and the lower the temperature, the more expected rewards affect the probability. For a low temperature (τ → 0 + {\displaystyle \tau \to 0^{+}}), the probability of the action with the highest expected reward tends to 1.